A RESEARCH PROJECT PROPOSAL OF A DIGITAL BUS BOOKING SYSTEM

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INTRODUCTION

Uganda's public bus system faces inefficiencies, including limited online booking and the need for passengers to visit bus parks, causing delays and inconvenience, especially in congested Kampala. Our solution is a mobile and web platform enabling trip booking, real-time bus tracking, and route management to improve efficiency

BACKGROUND

Public bus transportation is vital to Uganda's economy but hindered by outdated systems like physical ticketing and limited digital tools for bookings, tracking, and route management. Existing solutions focus on cross-border transport, neglecting intercity systems. A comprehensive digital platform offering online ticketing, real-time tracking, and route optimization is needed to boost efficiency and passenger convenience.

PROBLEM STATEMENT

Uganda’s public bus transport system suffers from inefficiencies in passenger convenience and service management. Passengers must travel to central bus parks even when they live along the bus route, causing unnecessary travel, congestion, and delays. Furthermore, limited use of a digital platform for bookings or real-time bus schedules creates uncertainty in travel planning and reduces service efficiency.

PROPOSED SOLUTION

Our proposed solution introduces a comprehensive mobile and web platform that will allow passengers to book trips, track buses in real time, and manage routes more effectively. By addressing the current gaps in service, the project aims to improve user convenience, optimize bus operations, and improve overall travel experiences across Uganda.

RESEARCH OBJECTIVES

General Objective: To develop a digital information system that improves Uganda’s public bus transportation system efficiently.

Specific Objectives

- To understand the gaps in the current Uganda's bus transport systems and gather requirements needed from stakeholders.
- To design a user-friendly web and mobile applications for Uganda’s public bus system.
- To develop the system by implementing the identified features using appropriate software development tools and technologies, ensuring user-friendliness and efficiency.
- To test and evaluate the system by conducting trials with real users to assess its performance, reliability, and user satisfaction, and necessary improvements before full deployment
- To deploy the system for use.

RESEARCH SIGNIFICANCE

Society: The digital bus booking system will enhance community convinience by allowing passengers to book tickets in advance, reducing wait times and improving travel planning.

Academia: .This research will offer valuable insights that can inform future researchers and provide practical recommendations on how to implement a digital bus booking system in the transportation industry.

Transportation industry: This project will enhance the transportation industry through streamlining processes, reducing manual errors and increasing operational efficiency through real-time scheduling.

RESEARCH SCOPE

Our scope covers the technological boundaries, geographical boundaries, and the time constraints of of the the digital bus booking system.

Technical Scope: The proposed system will include both a web and a mobile app, developed using technologies like Flutter and Laravel, and will be integrated with MTN and Airtel Mobile Money APIs.

Geographical Scope: The system will serve public transit users in Uganda most especially in cities like Kampala, Entebbe, Jinja, Mbarara, and Gulu, Wakiso, with potential expansion into other East African countries.

RESEARCH SCOPE

Time Scope: The expected timeline for the project is 8 Months as seen below;

1 Month	Requirements Gathering & Analysis
1 Month	System Design
4 Months	System Development
1 Month	Testing & Validation
1 Month	Deployment

LITERATURE REVIEW

The Literature Review compels existing literature related to the development and design of digital ticketing and bus booking systems. The review focuses on the key aspects of these systems, including their features, underlying technologies and the benefits and challenges associated with them.

Advantages

- Provides convenience and saves time
- Reduces paper work and operational costs
- It is easily scalable allowing easy expansion
- It improves operational efficiency through streamlined management of processes

Disadvantages

- Limited accessibility for users in rural areas
- May involve system downtimes
- Technology resistance
- They tend to be expensive

COMPARISONS OF EXISTING SYSTEMS & OUR PROPOSED SYSTEM

	Feature	Ugabus	Kacyber	Modern Coast	Trinity	Tiketi (for Volcano, Taqwa, Mash Poa)	Easy Coach	Proposed System
1	Real-time Seat Availability and Visualization	YES	YES	YES	YES	NO	NO	YES
2	Mobile-Friendly Interface	YES	YES	NO	YES	YES	YES	YES
3	Multiple Payment Options	YES	YES	NO	NO	YES	YES	YES
4	Trip Notifications	NO	YES	NO	YES	NO	YES	YES
5	User Booking Modifications	NO	YES	YES	NO	NO	YES	YES
6	Chatbot Support, Customer Support (24/7)	YES	YES	NO	NO	NO	NO	YES
7	Refund Processing	YES	NO	NO	NO	NO	YES	YES
8	Bus Tracking (Real-time)	NO	YES	NO	NO	NO	NO	YES
9	Route/ Trip Optimization	NO	YES	NO	NO	NO	NO	YES
10	Predictive Bus Arrival Times	NO	NO	NO	NO	NO	NO	YES
11	Dynamic Bus Scheduling	NO	NO	NO	NO	NO	NO	YES
12	Rating and Feedback mechanism	NO	NO	NO	NO	NO	NO	YES
13	Passenger Demand Forecasting & Analytics	NO	NO	NO	NO	NO	NO	YES
14	Real-time Traffic Integration	NO	NO	NO	NO	NO	NO	YES
15	Luggage tracking and passenger manifest	NO	NO	NO	NO	NO	NO	YES

METHODOLOGY

The methodology consists of the approach and methods that will be employed in the development of the digital bus booking system. It details how each specific objective will be achieved throughout the project's lifecycle, ensuring the final product meets the defined requirements.

Requirements Gathering

System Design

System Development

Testing & Validation

System Deployment